Customer Segmentation Using RFM Analysis in E-Commerce

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Problem Statement and Objective

Problem Statement:

E-commerce businesses generate vast amounts of transactional data but often struggle to leverage it for actionable customer segmentation. Understanding purchasing patterns is crucial for predicting behavior, improving retention, reducing churn, and maximizing customer lifetime value. A structured approach is needed to identify high-value customers, detect trends, and optimize targeted marketing strategies.

Objective:

- Utilize the RFM (Recency, Frequency, Monetary) framework to segment e-commerce customers.
- Analyze purchasing patterns to identify high-value customers and at-risk segments.
- Create targeted marketing strategies to improve customer retention and satisfaction.

Dataset Overview

- **Source:** [Kaggle - E-commerce Dataset]
- **Size:** (541909, 8)

Key Variables:

- **InvoiceNo:** Unique transaction ID
- **StockCode:** Unique product identifier
- **Description:** Product description
- **Quantity:** Number of items purchased
- **InvoiceDate:** Timestamp of purchase
- **UnitPrice:** Price per unit in GBP
- **CustomerID:** Unique customer identifier
- **Country:** Location of the customer

Data Cleaning and Preparation

1. Handled Missing Values
2. Removed Duplicates
3. Removed Cancelled Orders
4. Removing non-product Stock-Codes

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```
df.isnull().sum()

InvoiceNo           0
StockCode           0
Description      1454
Quantity            0
InvoiceDate         0
UnitPrice           0
CustomerID     135080
Country             0
```

The percentage of missing values in the CustomerID column is **24.93%**.

Since the analysis will revolve around investigating customers and clustering them into categories, the missing values in the CustomerIDs were removed.

Data Cleaning and Preparation

1. Handled Missing Values
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```
# Check for duplicate rows
duplicates = df.duplicated()

# Count the number of duplicate rows
print(duplicates.sum())
```

```
5225
```

The number of duplicate rows in the dataset is **5525**.

These rows were removed from the dataset.

Data Cleaning and Preparation

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4. Removing non-product Stock-Codes

```python
# Exploring the rows for which quantity is less than 0
df[df["Quantity"] < 0]
```

	InvoiceNo	StockCode	Description	Quantity
141	C536379	D	Discount	-1
154	C536383	35004C	SET OF 3 COLOURED FLYING DUCKS	-1
235	C536391	22556	PLASTERS IN TIN CIRCUS PARADE	-12
236	C536391	21984	PACK OF 12 PINK PAISLEY TISSUES	-24
237	C536391	21983	PACK OF 12 BLUE PAISLEY TISSUES	-24
...	...	...	...	...
401159	C581490	23144	ZINC T-LIGHT HOLDER STARS SMALL	-11
401243	C581499	M	Manual	-1
401410	C581568	21258	VICTORIAN SEWING BOX LARGE	-5
401411	C581569	84978	HANGING HEART JAR T-LIGHT HOLDER	-1
401412	C581569	20979	36 PENCILS TUBE RED RETROSPOT	-5

8872 rows × 8 columns

There are **8872** rows for which the quantity is negative which can be either due to data-entry errors or return orders or cancelled orders.

If we look at the InvoiceNo for all these cases, they start with the letter 'C' which indicates they are cancelled orders. Thus these rows were removed from the dataset

Data Cleaning and Preparation

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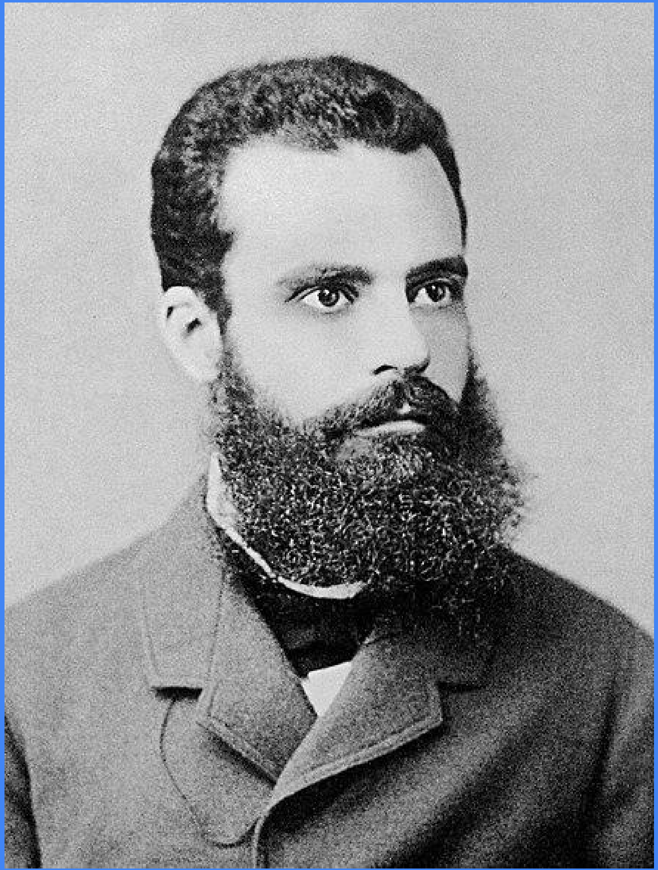
StockCodes	Description
POST	POSTAGE
C2	CARRIAGE
M	MANUAL
DOT	DOTCOM POSTAGE
BANK CHARGES	BANK CHARGES

There are certain StockCodes which do not belong to any products. All the rows containing such StockCodes were removed.

Pareto Principle

Roughly 80% of outcomes stem from 20% of causes



26%

Customers contribute to 80% of the revenue.

21%

Products contribute to 80% of the revenue.

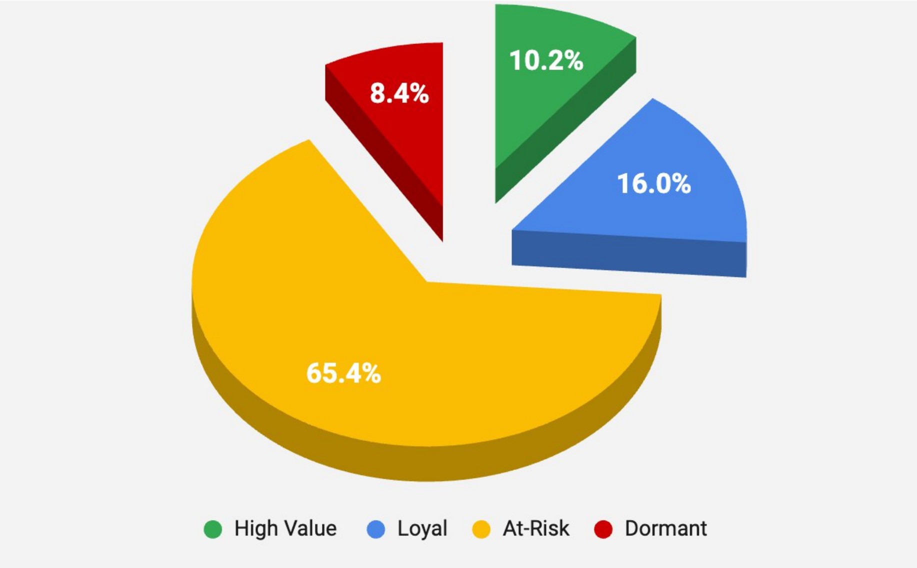
RFM Analysis and Customer Segmentation

What is RFM?

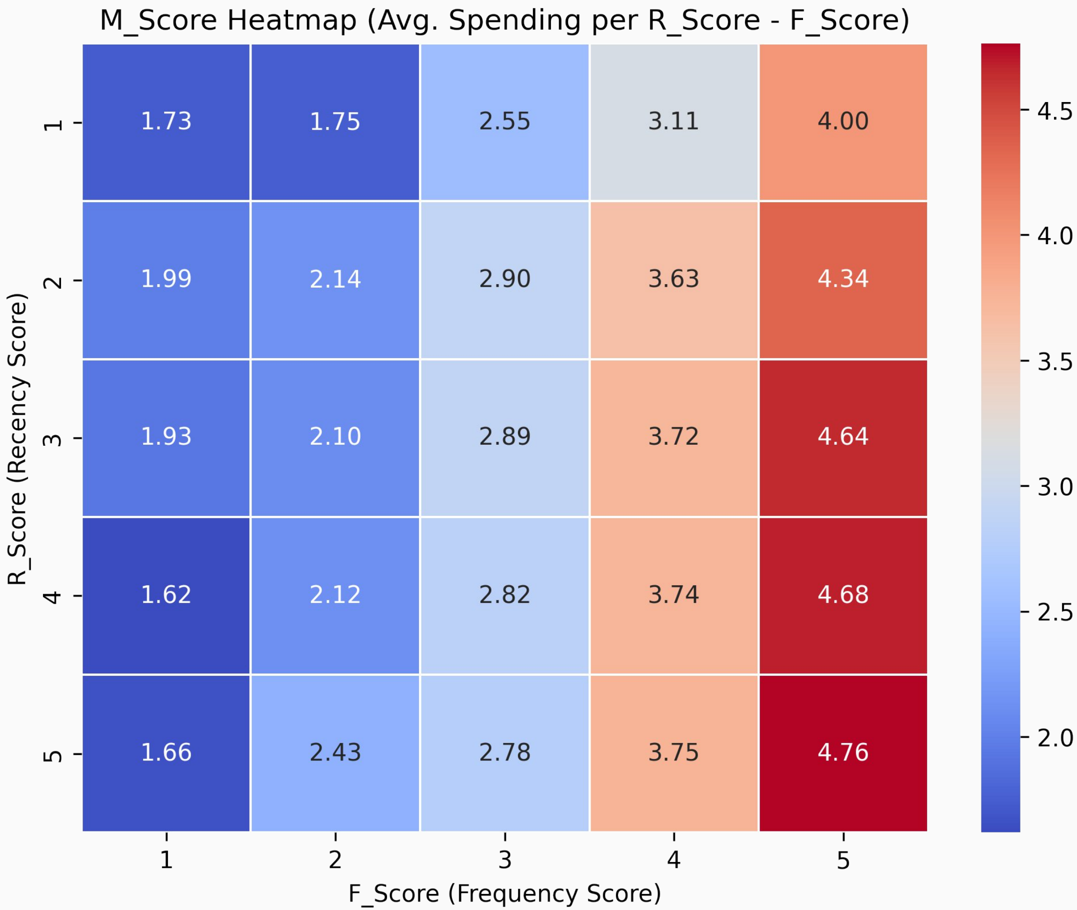
Recency (R): Days since last purchase
Frequency (F): Number of purchases
Monetary (M): Total spending

Customers are segmented into **five equal buckets** based on Recency, Frequency, and Monetary values. Each customer is ranked for each metric, assigned a **score from 1 to 5**, and their scores are summed to derive an overall **RFM score** for analysis.

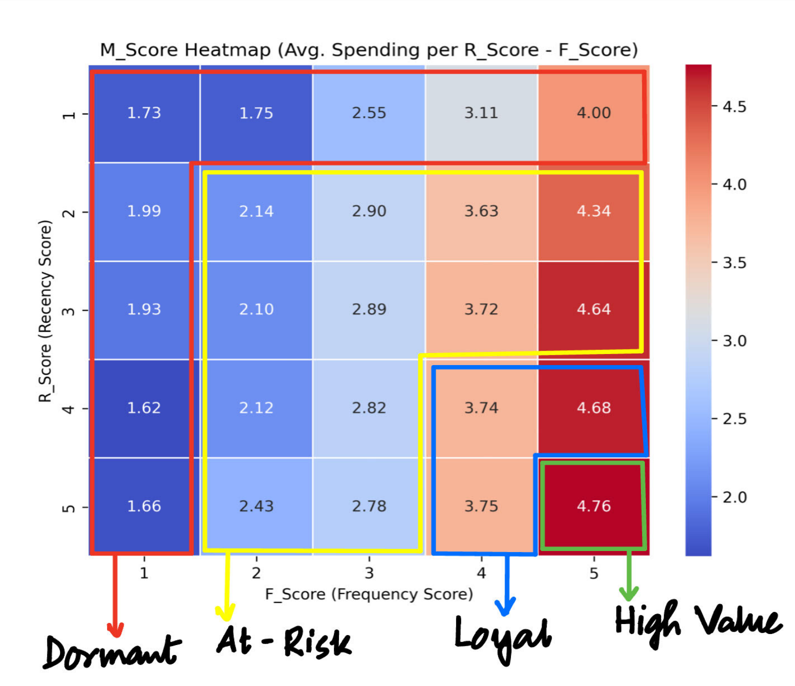
Customer Segments Based on RFM Score



Heat Map based on RFM Score



RFM Analysis and Customer Segmentation



Based on this RFM Score Analysis, we have four customer segments:

1. **High Value Customers**
2. **Loyal Customers**
3. **At-Risk Customers**
4. **Dormant Customers**

Dashboard

Country
All
Month
All
8.74M
Total Revenue
4335
Active Customers
5M
Sales Volume
Qtr 1
Qtr 2
Qtr 3
Qtr 4
2010
2011
RFM Heatmap
1 2 3 4 5
1 1.73 1.75 2.55 3.11 4.00
2 1.99 2.14 2.90 3.63 4.34
3 1.93 2.10 2.89 3.72 4.64
4 1.62 2.12 2.82 3.74 4.68
5 1.66 2.43 2.78 3.76 4.76
Revenue Contributions by Segment
Segment At-Risk Dormant High Value Loyal
1.0M
0.5M
0.0M
Jan 2011
Mar 2011
May 2011
Jul 2011
Sep 2011
Nov 2011
Customers by Segment
Segment At-Risk Loyal High Value Dormant
8.35%
10.22%
15.99%
65.44%
Revenue by Days of Week
2M
1M
0M
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Active Customers Over Time
2K
1K
Jan 2011
Apr 2011
Jul 2011
Oct 2011
Top/Bottom Customers
CustomerID TotalSales SalesVolume
14646 2,79,138.02 197420
18102 2,59,657.30 64124
17450 1,94,390.79 69973
16446 1,68,472.50 80997
14911 1,36,161.83 80404
12415 1,24,564.53 77669
14156 1,16,560.08 57755
Total 87,37,227.64 5155661

Recommendations Based on Segments

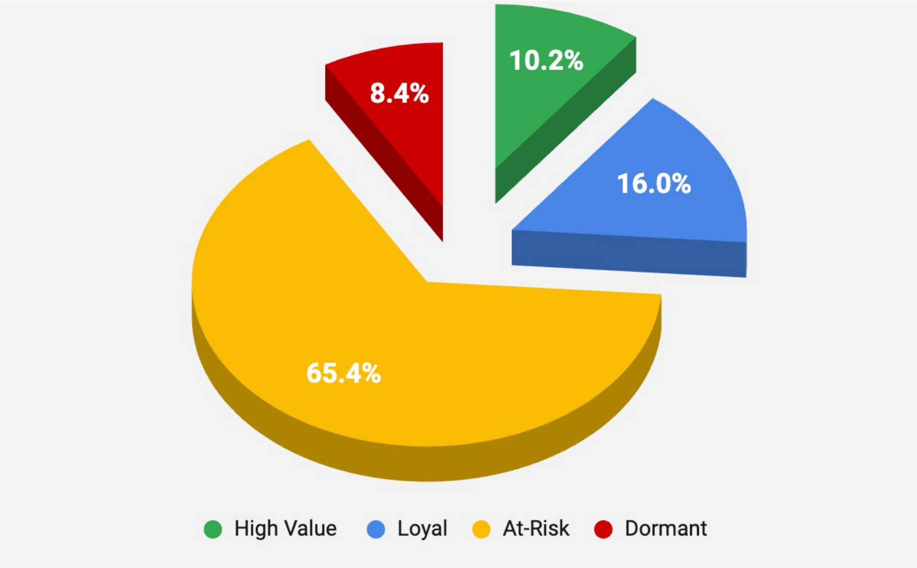
High Value Customers

Characteristics: Brand advocates with exceptional engagement and spending.

Behaviour: Low Recency (recent purchases), High Frequency, High Monetary.

Strategy: Reward with loyalty programs and exclusives.

Customer Segments Based on RFM Score



Recommendations Based on Segments

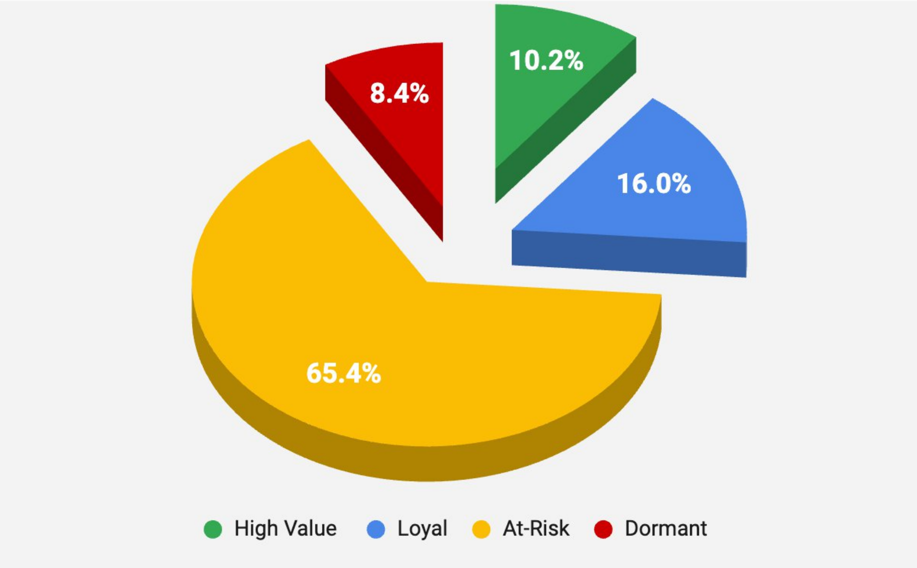
Loyal Customers

Characteristics: Consistent buyers with moderate activity.

Behaviour: Average Recency, Moderately High Frequency, High Monetary.

Strategy: Upsell/cross-sell opportunities to boost value.

Customer Segments Based on RFM Score



Recommendations Based on Segments

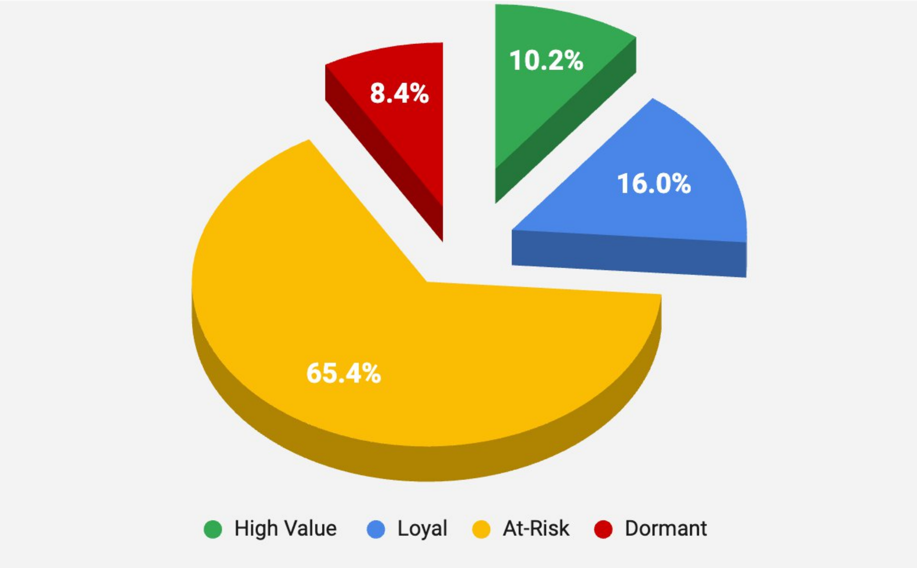
At-Risk Customers

Characteristics: Not-so Consistent buyers

Behaviour: Moderately High Recency, Low Frequency, Variable Monetary.

Strategy: Personalized emails and Limited-time promotions to regain interest.

Customer Segments Based on RFM Score



Recommendations Based on Segments

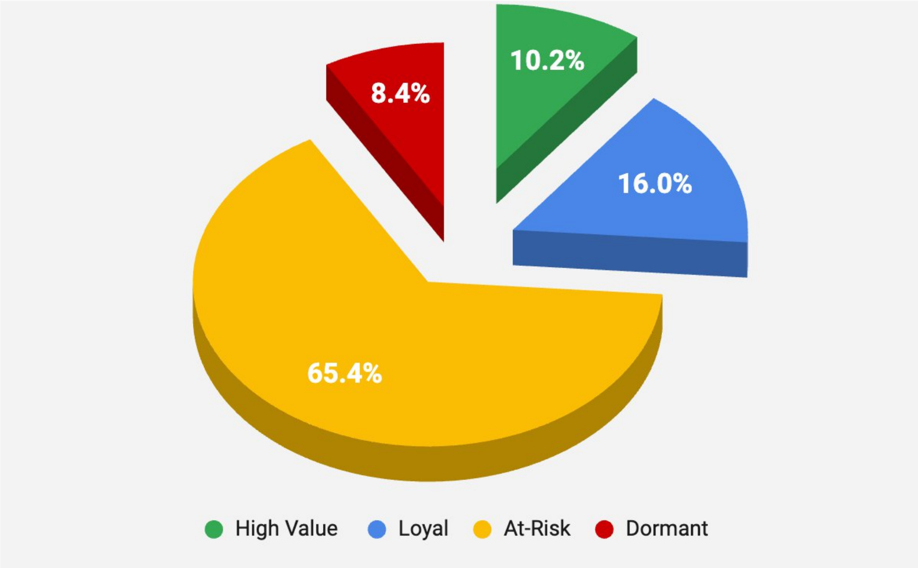
Dormant Customers

Characteristics: Declining engagement with potential churn risk.

Behaviour: High Recency, Below Average Frequency, Low Monetary.

Strategy: Win-back campaigns or surveys to address dissatisfaction.

Customer Segments Based on RFM Score



Thanks!

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